IT3091 Machine Learning - Personal Learning Journey

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My responsibility was the weather, location, and time-preparation stage of the group workflow. I learned that a weather variable is useful only when its location and timing fit the decision being modelled; otherwise, an apparently valid feature can give a misleading result.

I joined weather location IDs to the supplied location table instead of guessing place names, then prepared the daily weather measurements for analysis. The group workflow used rainfall, mean temperature, wind speed, solar radiation, and reference evapotranspiration. Location checks helped keep the geographic meaning of the records explicit.

I helped define prior three-month weather windows for the earlier seasonal planning analysis: January to March before the Yala planning date and June to August before the Maha planning date. The workflow checked that the dates were complete and that each weather window ended before the forecast date. Twenty-four season-year windows passed those checks in the documented common period.

The current public app has since been organized around two separate pipelines: weekly Colombo wholesale price and district/season crop yield. The prepared weather work remains part of the project's group evidence, but weather fields are not inputs to those current deployed models. Recognizing that distinction taught me that feature preparation and final model selection are separate decisions, and that a report must say which features the running app actually uses.

If I extended this work, I would compare district-level weather with national summaries and evaluate whether local coverage is complete enough for farm planning. My main learning was to make time windows testable, document what each feature represents, and avoid using information that would not be available at the decision date.